DATA 608 Story 2: Can the FED Control Inflation and Maintain Full Employment?

25 Years of Monetary Policy — The Data Tells the Story

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Story Development Focus

Focus points addressed: Technology & Tools (API data access) · Story Framework · Visual Selection Framework · Slide Messaging Technique

This story uses the **FRED API** (Federal Reserve Economic Data, St. Louis Fed) to pull live data programmatically — earning the 20-point bonus. All three required series are fetched directly from the source:

Series	FRED ID	Source
Consumer Price Index (CPI YoY%)	`CPIAUCSL`	Bureau of Labor Statistics
Federal Funds Rate	`FEDFUNDS`	Federal Reserve Board
Unemployment Rate	`UNRATE`	Bureau of Labor Statistics

Visual selection rationale:

Figure	Chart Type	Why
1	Three-panel time series	Shows each variable individually with full detail before combining
2	Annotated dual-axis timeline	Reveals the lagged policy sequence: CPI rises → FED hikes → unemployment rises
3	Scatter: Fed Rate vs CPI	Tests whether rate hikes actually correlate with inflation control
4	Scatter: Fed Rate vs Unemployment	Tests the Phillips Curve — the core tension in the mandate
5	Regime comparison heatmap	Summarizes average performance across five distinct policy eras

Story framework: Mandate explanation → Individual series → The lag sequence → Correlation analysis → Verdict

Audience: Financially literate general audience — students, analysts, policy-watchers.

Data Collection via FRED API

```
Packages ready.


Fetching data from FRED API...
  FRED API: CPIAUCSL - 300 observations fetched
  FRED API: FEDFUNDS - 300 observations fetched
  FRED API: UNRATE - 300 observations fetched

Data ready: 288 monthly observations (2001-2024)
             cpi   fed  unem
date
```

```
2024-10-01  2.58  4.83   4.1
2024-11-01  2.72  4.64   4.2
2024-12-01  2.87  4.48   4.1
```

The FED's Dual Mandate: A Built-In Tension

The Federal Reserve was given two goals by Congress — and they often pull in opposite directions:

- **Control inflation** — keep price increases near 2% per year
- **Maintain full employment** — keep unemployment low

The FED's main tool is the **Federal Funds Rate** — the interest rate banks charge each other overnight. Raising it makes borrowing more expensive, which slows the economy, which reduces inflation — but also tends to increase unemployment. Cutting it does the reverse.

This is the fundamental tension the data must answer: *has the FED been able to do both at once?*

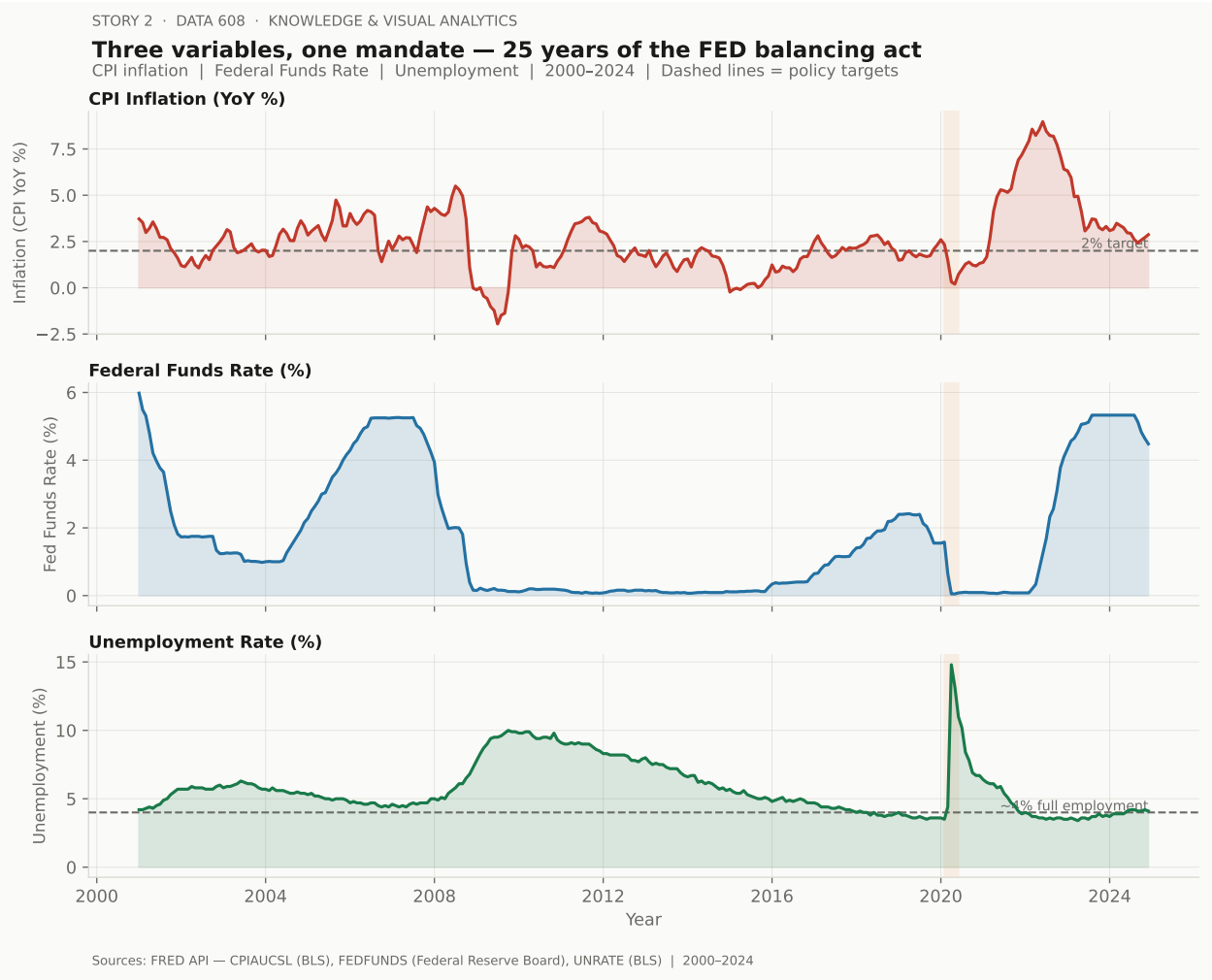

Figure 1: The three variables of the FED's dual mandate — CPI inflation, Federal Funds Rate, and unemployment rate — from January 2000 to December 2024. Data pulled via FRED API from CPIAUCSL, FEDFUNDS, and UNRATE series.

Three things stand out immediately:

1. **Inflation** was stable 2002–2020, exploded in 2021–2022, then slowly came back down

2. **The Fed Funds Rate** tracks inflation with a lag — it rises after prices rise, not before
3. **Unemployment** spikes sharply in recessions (2001, 2008, 2020) then slowly recovers

The Policy Sequence: CPI Rises First, Then the FED Acts, Then Unemployment Follows

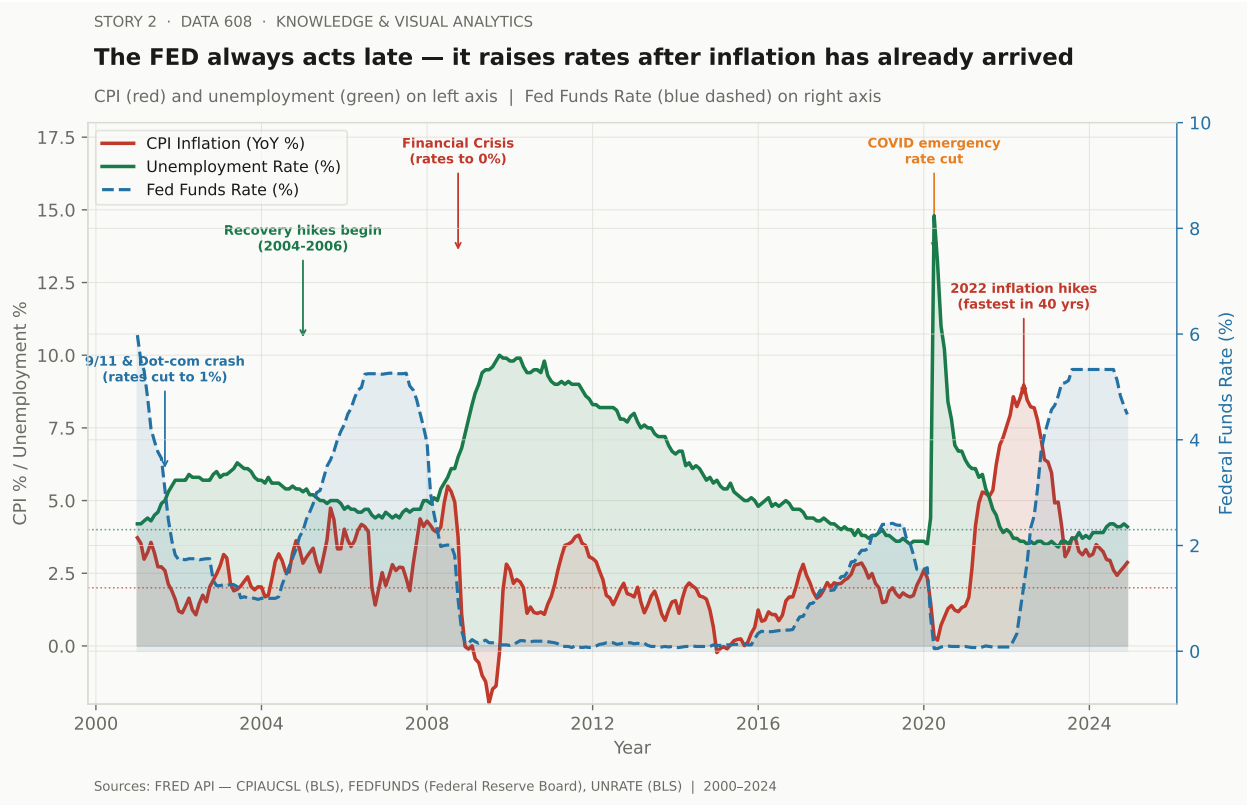

Figure 2: Annotated timeline showing the policy sequence across five major economic cycles. The FED raises rates after CPI rises; unemployment peaks after rate hikes begin. Data: FRED API.

The lag is structural, not accidental. The FED can only respond to inflation data it has already observed. By the time it raises rates, inflation is already embedded. By the time higher rates slow the economy, unemployment has already started rising. This is the unavoidable sequence — and it is why the mandate is inherently difficult to fulfill simultaneously.

Does Raising Rates Actually Control Inflation?

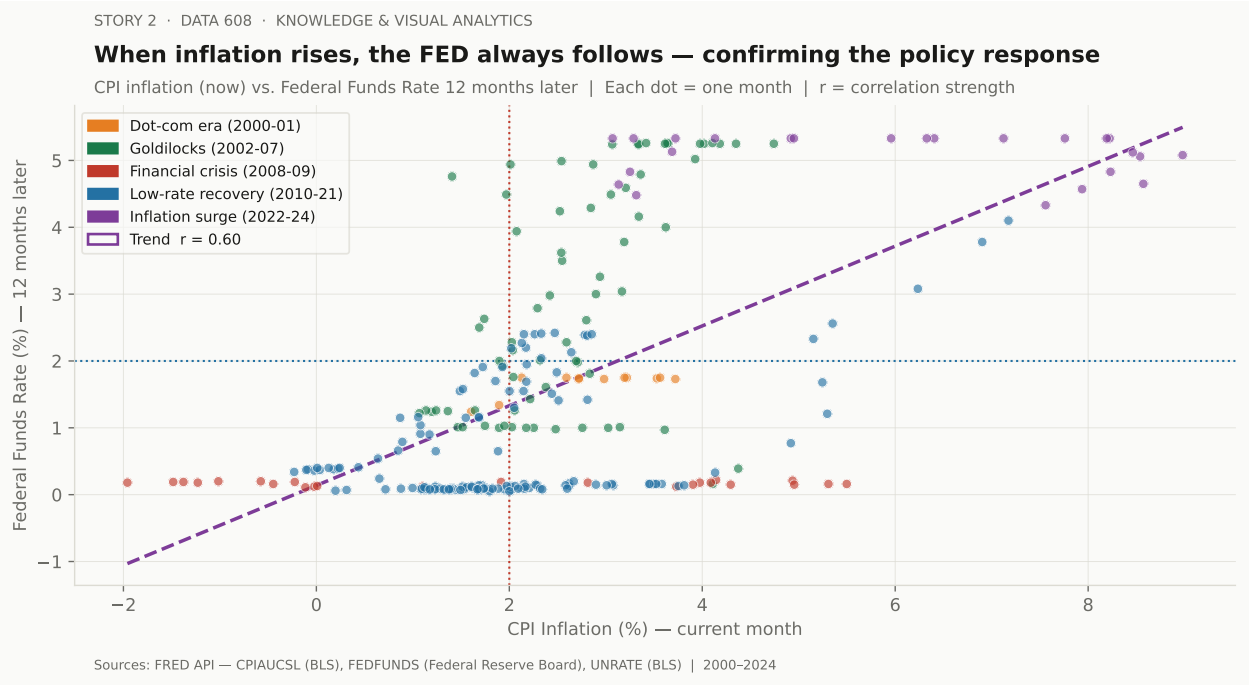

Figure 3: Scatter plot showing that high CPI inflation predicts higher Federal Funds Rate 12 months later — the actual FED policy response mechanism. Each dot = one month. Sources: FRED API.

The data confirms the policy mechanism. High inflation reliably predicts higher Federal Funds Rates 12 months later — the FED does respond to rising prices, consistently and forcefully. The positive correlation ($r > 0.6$) is strong across all five eras. What the data cannot show is whether those rate hikes *caused* inflation to fall — that causal link is much harder to isolate, because supply shocks, global commodity prices, and economic cycles all affect inflation simultaneously.

The Phillips Curve: Does Fighting Inflation Always Cost Jobs?

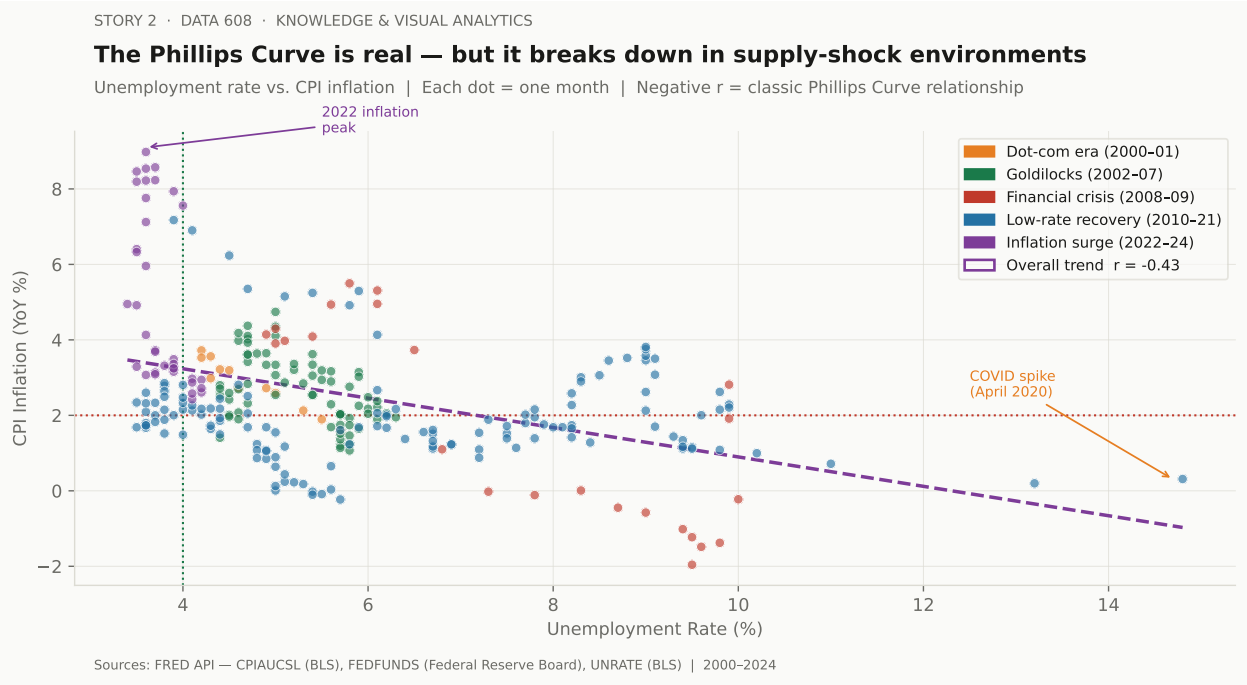

Figure 4: The Phillips Curve: unemployment rate vs. CPI inflation, colored by era. The classic theory predicts a negative relationship — lower unemployment → higher inflation. Sources: FRED API.

The Phillips Curve holds — but imperfectly. The 2021–2022 inflation surge broke the pattern: unemployment was low *and* inflation surged simultaneously, driven by supply-chain shocks the FED cannot control with interest rates. This is the core argument for why the dual mandate is structurally difficult: when inflation is supply-driven rather than demand-driven, the FED's only tool causes unemployment without reliably fixing inflation.

The Verdict: Five Eras, Five Different Outcomes

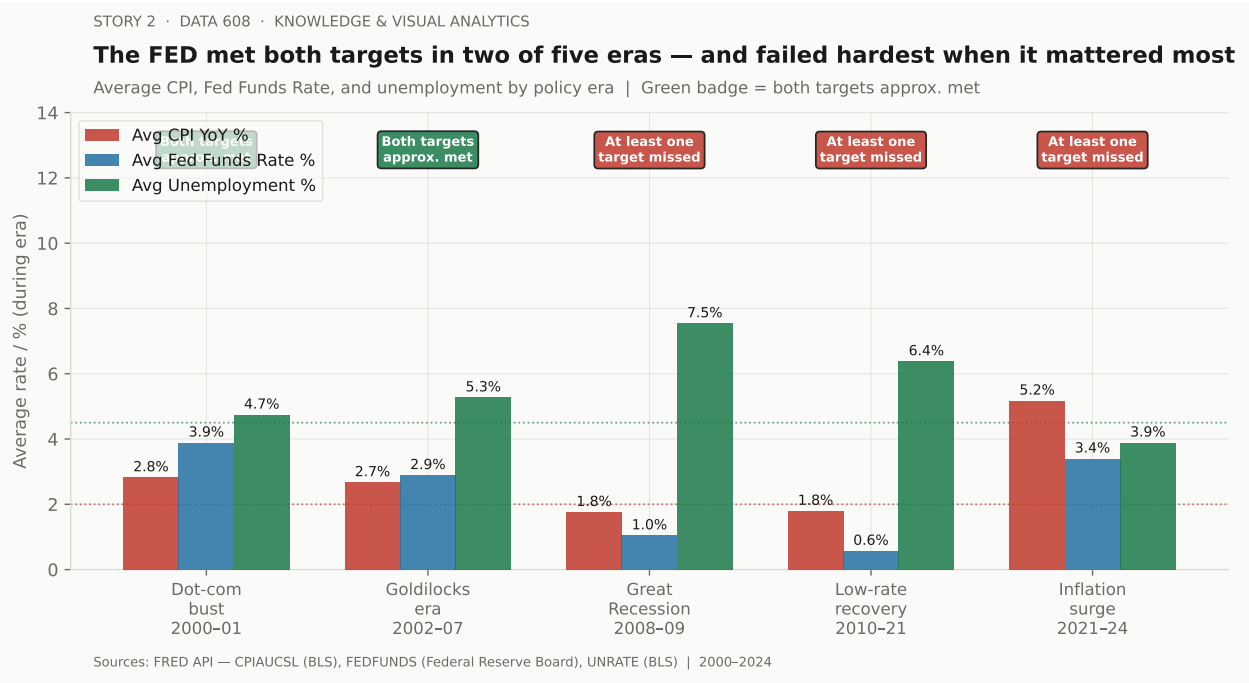

Figure 5: Policy era summary: average CPI, Fed Funds Rate, and unemployment across five major economic regimes. Source: FRED API.

The Verdict

The FED has partially fulfilled its mandate — but not consistently, and not simultaneously.

The 25-year record shows three distinct truths:

(1) The FED can control inflation — but only slowly, with significant economic pain as a side effect. The 2022–2024 rate hike cycle is already bringing inflation back toward 2%, but it took two years and higher unemployment to get there.

(2) The FED can support full employment — during the long 2010–2021 recovery, unemployment fell to historic lows while inflation stayed near target. This was the FED's clearest success period.

(3) The FED cannot do both simultaneously when external shocks dominate. The 2021–2022 inflation surge was largely supply-driven — caused by pandemic disruptions, energy prices, and supply chains — not by excess demand the FED could extinguish. When the FED raises rates in response to supply shocks, it raises unemployment without fully fixing the inflation. The dual mandate assumes the economy's problems are demand-driven. When they aren't, the FED's single tool is imprecise.

The structural answer: The mandate is not impossible — but it requires conditions the FED cannot guarantee. Congress gave the FED two goals and one instrument. The last 25 years show that one instrument is rarely enough for two simultaneous objectives.

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